Comparative Study of 2D Lid-Driven Cavity Flow Across Python, Ansys Fluent, and OpenFOAM

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Introduction

The two-dimensional lid-driven cavity problem has long been considered a benchmark case in computational fluid dynamics, extensively studied and validated both experimentally and numerically. With the advancement of computational power, obtaining numerical solutions to this problem has become relatively accessible even with only fundamental knowledge of fluid dynamics. In this study, a Python code based on the projection method is developed to solve the two-dimensional lid-driven cavity, and the results are compared against the classical data of Ghia et al., commercial software (Ansys), and the open-source solver (OpenFOAM).

Lid Driven Cavity

◆ **Solve 2D lid driven cavity flow with Numerical Analysis**

<Governing Equation>

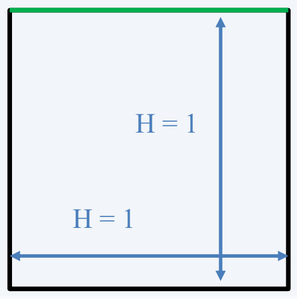
• Navier-Stokes equations

$$\frac{\partial \boldsymbol{u}}{\partial t} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} - (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$$

• Continuity equation

$$\nabla\cdot\boldsymbol{u} = 0$$

$u_{lid}(t \geq 0) = U, v = 0$



$u(\boldsymbol{x}, t < 0) = 0$
$v(\boldsymbol{x}, t < 0) = 0$

$u = v = 0$

Reynolds Number

$$Re = \frac{u_{lid}H}{\nu}$$

Python

<Numerical Modeling>

• Convection term
  2nd Adams-Bashforth method

• Diffusion term
  Crank-Nicolson method

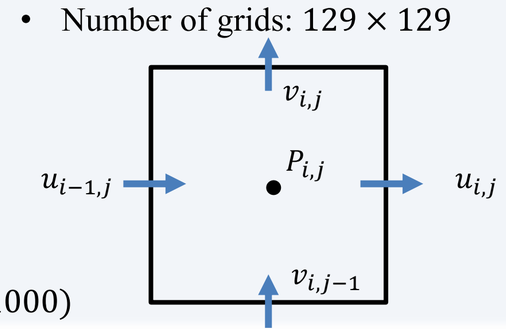
• Projection method

• $\Delta t = 0.01 (Re = 100, 400), 0.005\ (Re = 1000)$

<Spatial Grids>

• Staggered grids

• Number of grids: 129×129



Ghia et al (1982)

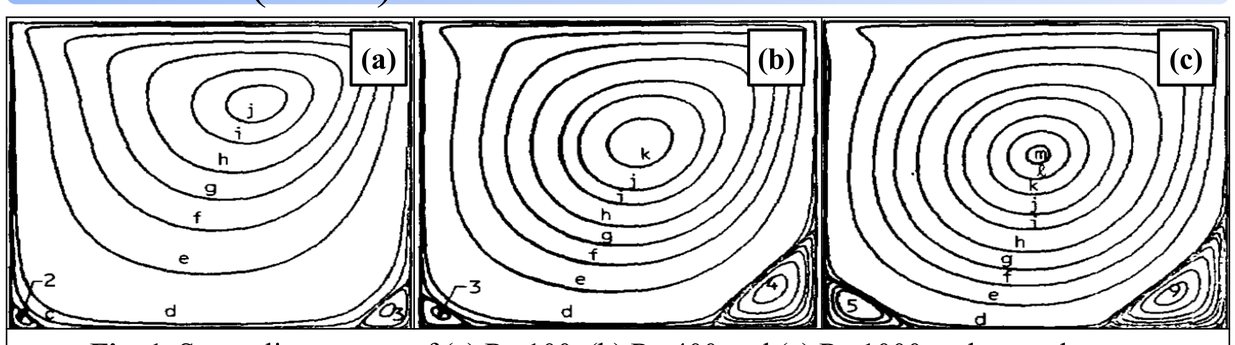


Fig. 1 Streamline pattern of (a) Re 100, (b) Re 400 and (c) Re 1000 at the steady state

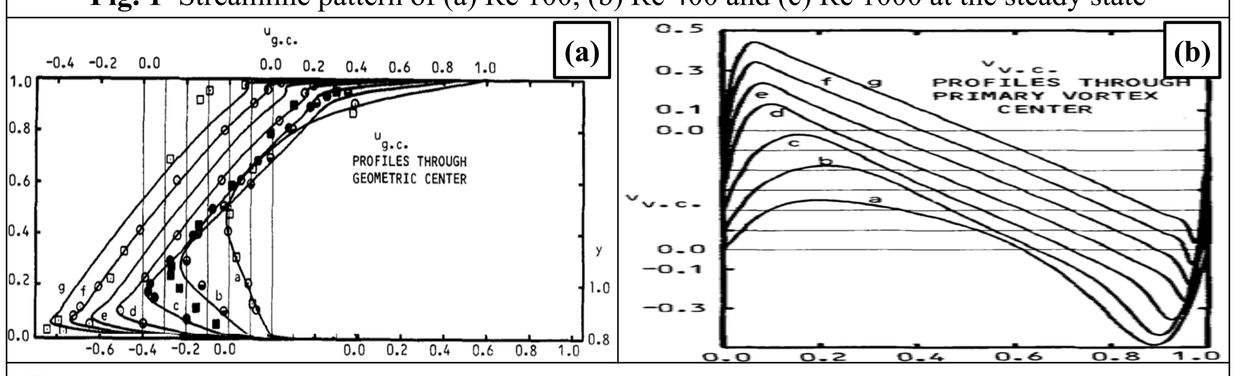


Fig. 2 Comparison of (a) u-velocity along vertical lines and (b) v-velocity along horizontal lines

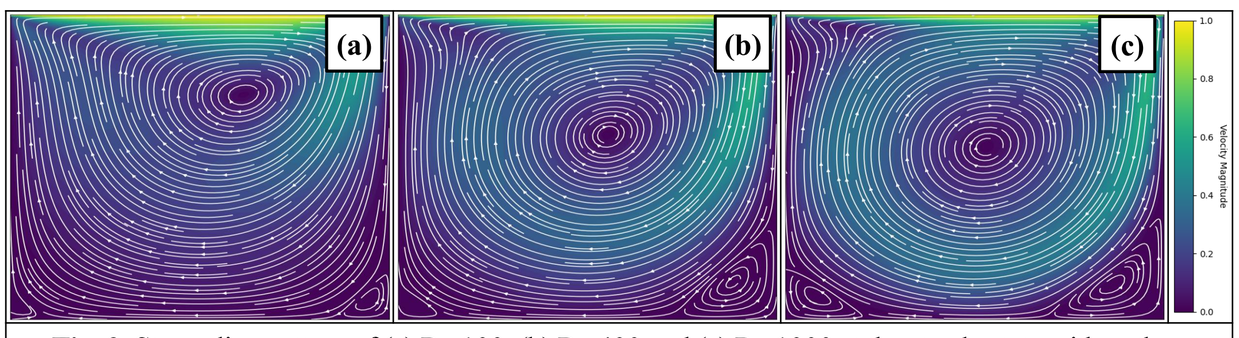


Fig. 3 Streamline pattern of (a) Re 100, (b) Re 400 and (c) Re 1000 at the steady state with python

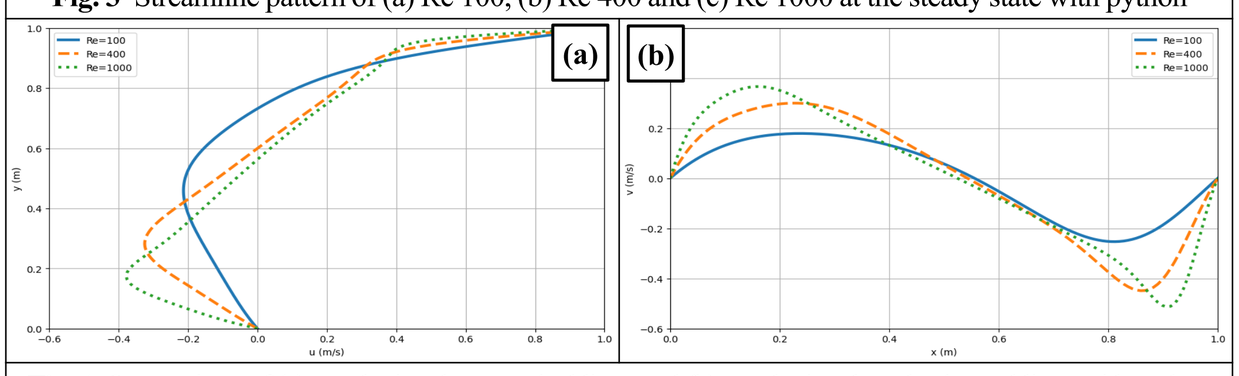


Fig. 4 Comparison of (a) u-velocity along vertical lines and (b) v-velocity along horizontal lines with python

Ansys Fluent

ANSYS is a commercial software package capable of simulating a wide range of engineering problems.

In this study, ANSYS Fluent 2025 R1 was used, and the computations were performed using the Finite Volume Method (FVM).

<Numerical Modeling>

• Pressure-Based Solver

• SIMPLE(Semi-Implicit Method for Pressure-Linked Equations) scheme

• Convection term
  Second order Upwind

• Diffusion term
  Central differencing (second order, default)

• Transient Formulation
  Second order Implicit

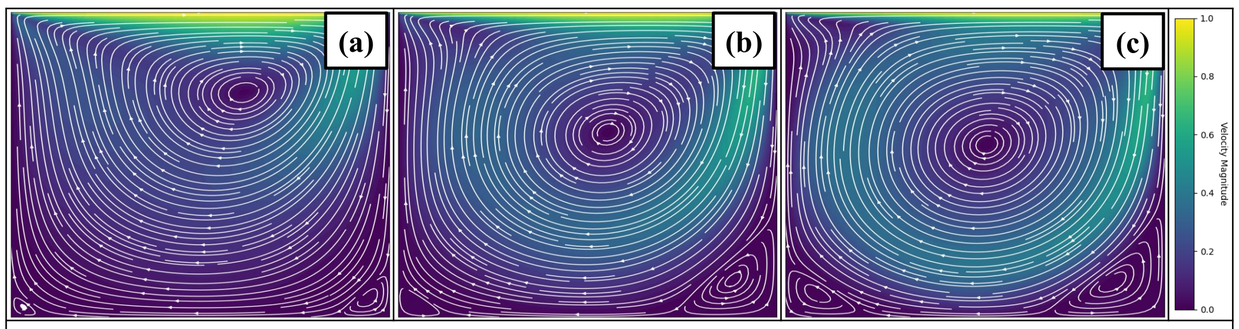


Fig. 5 Streamline pattern of (a) Re 100, (b) Re 400 and (c) Re 1000 at the steady state with Ansys

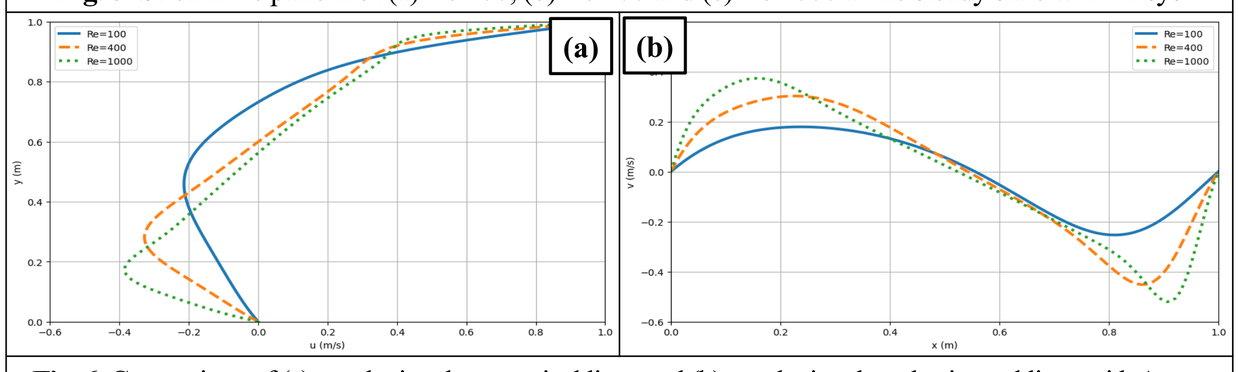


Fig. 6 Comparison of (a) u-velocity along vertical lines and (b) v-velocity along horizontal lines with Ansys

OpenFOAM (**Open**-source **F**ield **O**peration **A**nd **M**anipulation)

OpenFOAM is a free open-source CFD software which written in C++. It primarily solves problems in three dimensions.

However, by specifying the z-direction thickness as one cell layer and setting the boundary conditions 'empty' for the front and back faces, it can perform 2D simulations.

<Numerical Modeling>

• Explicit operation
  Finite Volume Calculus

• Implicit operation
  Finite Volume Method

• PISO (Pressure-Implicit with Splitting of Operators) algorithm

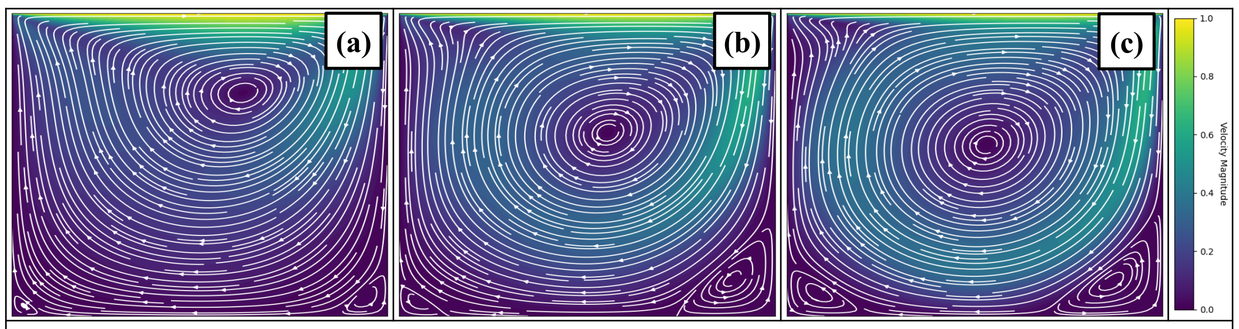


Fig. 7 Streamline pattern of (a) Re 100, (b) Re 400 and (c) Re 1000 at the steady state with OpenFOAM

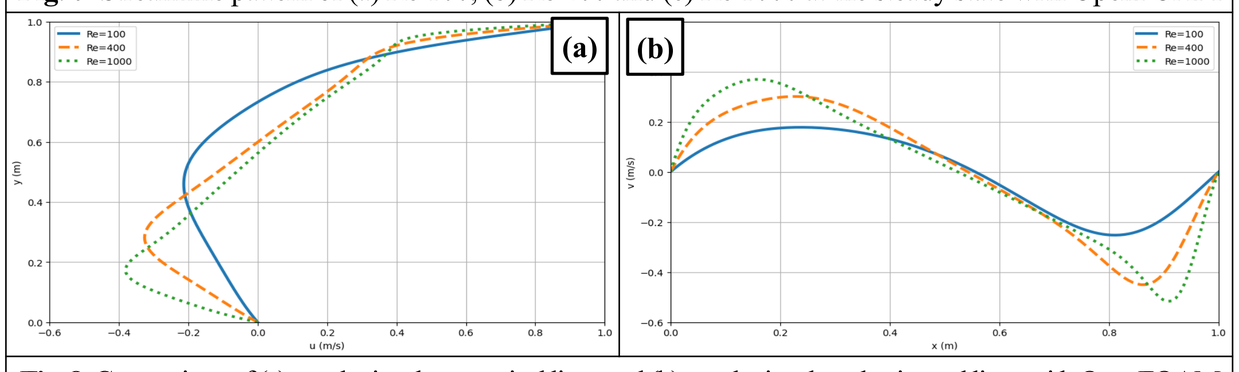


Fig. 8 Comparison of (a) u-velocity along vertical lines and (b) v-velocity along horizontal lines with OpenFOAM

Conclusion

This study applied a Python projection method, Ansys, and OpenFOAM to the two-dimensional lid-driven cavity problem. The results closely matched the benchmark data of Ghia et al. and showed strong agreement among the methods, confirming their reliability for this classical CFD case. Furthermore, the approach can be extended to rectangular cavities with different aspect ratios and is also applicable to three-dimensional configurations.